

Draft - expert review pending

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Topic 4.1 - Cyber threats and how to stay safe

A real classroom situation

During exam season, the head teacher of a Karachi primary school receives a message that looks exactly like the education department's: the logo, the official tone, a link to "view your school's revised exam schedule". She taps the link, types the portal password, and the page shows an error. Nothing else happens - visibly. Two days later the school's portal account is locked and a message arrives asking for money to unlock it. She had been phished: the page she typed her password into belonged to nobody in the department. The teachers who later examined the message found three small tells - the sender's address was wrong by one letter, the greeting was generic, and the link's address, read carefully, was not the department's at all. This topic teaches you to see those tells before you tap.

The three families of cyber threats

Most attacks are cast wide; they catch the careless, not the chosen



Defences are habits: update, verify before you tap, install from official stores, back up away from the device.
Protecting the CIA triad: confidentiality, integrity, availability.

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Explanation

Common cyber threats: malware, phishing, social engineering

The threats that reach ordinary users come in three families. Malware is harmful software that gets onto a device to do damage or steal: a virus spreads by attaching itself to files and programs; ransomware locks your files and demands money to unlock them, exactly as in the head teacher's story; spyware sits quietly and reports what you type, including passwords. Malware usually arrives as an attachment or a download from a site you did not mean to visit - the "I Love You" virus of May 2000 spread through an email attachment and hit roughly 50,000 computers (Bourgeois et al., 2019).

Phishing is a message - email, SMS, or chat - that looks like it is from a trusted source but leads you to a fake page or a harmful link. The message is the bait; your password or your tap is the catch. Bourgeois and colleagues (2019) describe it plainly: phishing "occurs when a user receives an e-mail that looks as

if it is from a trusted source", drawing them to a site that captures their credentials.

Social engineering is the family that needs no software at all: the attacker simply talks or tricks. Bourgeois and colleagues (2019) give the classic case, pretexting: an attacker "calls a helpdesk or security administrator and pretends to be a particular authorized user" to get a password reset. A caller claiming to be "from the bank's head office, verifying your account" is the same move on a phone. The defence is not technical; it is a habit: verify who is asking before you answer.

Threats and the habit that stops each

Threat	How it reaches you	The practice that stops it
virus	infected attachments and downloads	do not open unexpected attachments; scan downloads
ransomware	phishing links and malicious downloads	keep software updated; back up files offline
spyware	fake apps and shady installs	install only from official stores
phishing	convincing fake messages	check sender address and link before tapping
social engineering	calls and messages that pressure you	verify the asker through a known channel

Every row: the habit is something a person does, not something a box does.

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Online security practices and defences

The good news is that a small set of habits stops most attacks, because most attacks are not targeted at you personally - they are cast wide, like nets, and catch only the careless. The CIA triad names what you are protecting: confidentiality, keeping information accessible only to those allowed (Bourgeois et al., 2019); integrity, "the assurance that the information being accessed has not been altered and truly represents what it intended"; and availability, information being reachable when authorised people need it.

The habits follow from the triad. **Update** your devices and apps: most updates close holes attackers already know about. **Verify before you tap**: read the sender's full address, be suspicious of urgency ("your account will be closed today!"), and reach officials through a channel you already trust rather than a link in a message. **Install only from official stores**, and be equally careful of unknown USB drives. **Back up** what matters, and keep one backup away from the device itself - Bourgeois and colleagues (2019) note that disasters often destroy backups stored in the same place as the originals. **Use antivirus**, but do not lean on it alone: it recognises known malware, not new tricks or old-fashioned talking. Pakistan's National Cyber Security Policy names this awareness layer explicitly, treating user awareness as part of the country's defence (Ministry of IT and Telecommunication, 2021), and the National Cybersecurity Alliance's guidance for everyday users is built on exactly these habits (National Cybersecurity Alliance, n.d.).

A caution against two common misconceptions. "My data is not worth stealing" is false: a school marks sheet, a parent-contact list, or simply your account itself (useful for sending more phishing in your name) is worth an attacker's while. And "antivirus software alone keeps me safe" is the more dangerous half-truth: the head teacher's portal was phished, and no antivirus stops a person typing a real password into a real-looking fake page. Nor is everyday confidence evidence of knowledge: a survey of 517 university students found weak cybersecurity awareness across malware, passwords and social media, despite lives lived online (Erendor and Yildirim, 2022). People, not only software, are the defence.

Activity: Spot the phish

Work in pairs. About 20 minutes.

1. Your teacher shows four messages (printed or projected), each supposedly from a school, a bank, a courier company, or the education department. One is genuine; three are phishing.
2. For each message, list every tell you can find: the sender's address, the greeting, the urgency, the link's real destination (described by the teacher), the request it makes.

3. Mark each message genuine or phishing, and write the single strongest piece of evidence for your verdict.
4. For one phishing message, write the reply you would actually send - or better, the action you would take instead of replying.

Check your understanding

1. Name the three families of cyber threats from this topic, with one example of each. (*Remember*)
2. Explain the difference between phishing and social engineering in two sentences. (*Understand*)
3. A message says: "Your account will be closed in 2 hours, click here to verify." List three tells that suggest phishing and the action you would take instead. (*Apply*)
4. Name the three parts of the CIA triad and say what each protects. (*Remember*)

Summary

The threats that reach ordinary users are malware (virus, ransomware, spyware), phishing, and social engineering - and most are cast wide, catching the careless rather than the chosen. The defences are habits, not gadgets: update, verify before you tap, install from official sources, back up away from the device, and treat unexpected human asks with the same suspicion as unexpected attachments. The CIA triad - confidentiality, integrity, availability - names what all of it protects.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the three threat families with an example of each.
- ☐ I can list at least three tells of a phishing message and the safe alternative to tapping.
- ☐ I can name the CIA triad and the habits that defend each part.

Try this at your practicum school

Ask the office how they receive official notices from the department (email, WhatsApp, portal). Write three sentences: which channel a phishing message could imitate, what the office currently checks before trusting a message, and one habit you would recommend.

Summative task

The head teacher asks you to prepare a one-page "safe messages" guide for the school's staff room: the three threat families in plain words, five tells of a phishing message, and the school's rule for what to do when a message asks for a password or money. Use one real-sounding example (invented by you) and mark its tells.

Marking guide (10 marks):

- The three threat families explained accurately and plainly: 3 marks.
- Five tells chosen well, each with a one-line explanation: 3 marks.
- The rule is actionable and the example's tells are marked correctly (Analyse): 3 marks.
- Fits one page, plain language for staff: 1 mark.

Further reading

- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 6. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Erendor, M. E., & Yildirim, M. (2022). Cybersecurity awareness in online education: A case study analysis. *IEEE Access*, 10, 52319-52335. <https://doi.org/10.1109/ACCESS.2022.3171829>
- National Cybersecurity Alliance. *Stay Safe Online*. <https://staysafeonline.org/>
- Ministry of Information Technology and Telecommunication. (2021). *National Cyber Security Policy*. Government of Pakistan. <https://moitt.gov.pk/>

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